

PCOS Status and Inflammatory Marker Levels Across the Menstrual Cycle

Session 2 presentation figures and tables

Table of contents

Research Question	1
Figure 1: Dataset Workflow	2
Table 1: Participant Characteristics	2
Menstrual-Cycle Time Points	3
Figure 2: Cytokine Comparison with Model-Based p-values	5
Results Interpretation	6
Conclusion	6

Research Question

Do inflammatory marker levels differ between participants with PCOS and participants without PCOS after accounting for menstrual-cycle stage?

Figure 1: Dataset Workflow

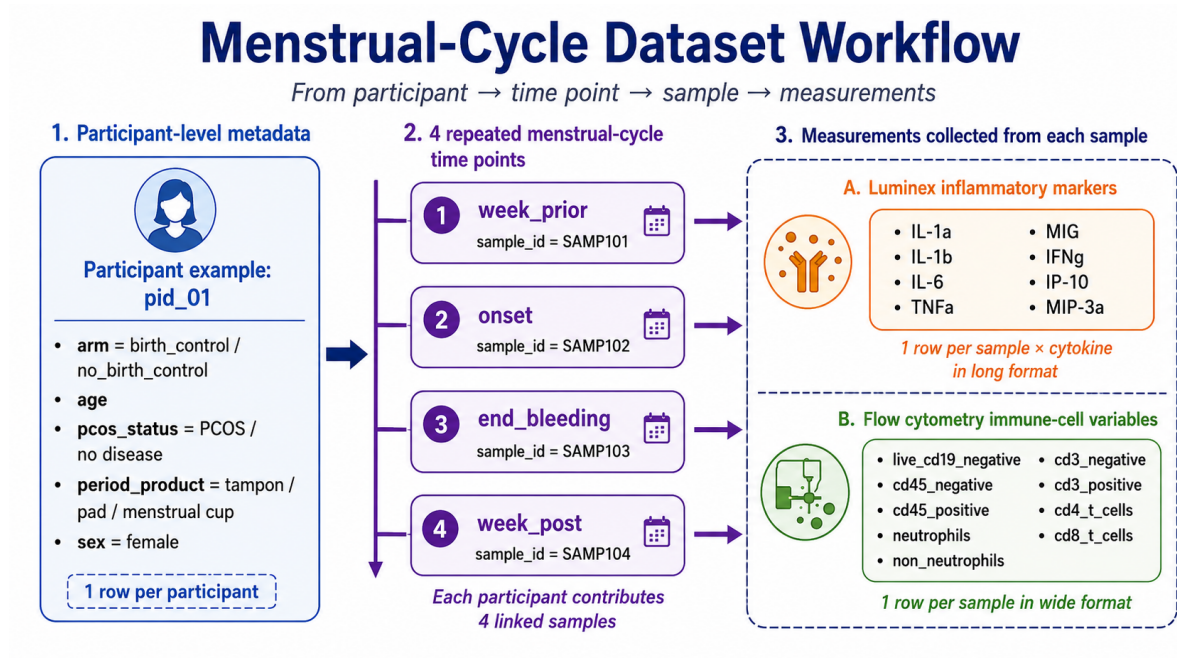


Figure 1. Structure of the menstrual-cycle dataset. Participant-level metadata are linked to four repeated menstrual-cycle samples per participant. Each sample is connected to Luminex inflammatory-marker measurements and flow-cytometry immune-cell measurements.

This figure introduces the data hierarchy: participant → time point → sample → measurements. Because each participant contributes repeated samples, samples from the same participant should not be treated as independent.

Table 1: Participant Characteristics

Characteristic	Overall (N = 27)	No disease (N = 13)	PCOS (N = 14)	p-value
Age (years), Mean (SD)	31.4 (2.9)	30.5 (2.5)	32.1 (3.1)	0.2 ^a
Birth control use, n (%)				0.2 ^b
birth_control	15 (56%)	9 (69%)	6 (43%)	

Characteristic	Overall (N = 27)	No disease (N = 13)	PCOS (N = 14)	p-value
no_birth_control	12 (44%)	4 (31%)	8 (57%)	0.2 ^c
Period product, n (%)				
menstrual_cup	4 (15%)	2 (15%)	2 (14%)	
pad	4 (15%)	0 (0%)	4 (29%)	
tampon	19 (70%)	11 (85%)	8 (57%)	

Note: All participants were recorded as female.

^a Wilcoxon rank sum test; ^b Pearson's chi-squared test; ^c Fisher's exact test.

Table 1. Participant characteristics by PCOS status.

Table 1 describes the participant-level cohort used for the PCOS versus no-disease comparison. The analysis includes 27 participants, with repeated samples collected across four menstrual-cycle time points.

Menstrual-Cycle Time Points

The four menstrual-cycle time points were ordered as: week_prior → onset → end_bleeding → week_post. Time point was included in the mixed-effects model to account for menstrual-cycle stage.

Figure 2: Cytokine Comparison with Model-Based p-values

Figure 1 — Cytokine Comparison with Model-Based p-values

Data Science for Biology Workshop

Data Import & Merge

Load Model Results

Figure 1 — Cytokine Comparison with BH-Adjusted p-values

```
# =====  
# Boxplots of log-transformed cytokine concentrations by PCOS status  
# Each panel annotated with Benjamini-Hochberg adjusted p-value  
# from the linear mixed-effects model  
# =====  
  
# Compute y-position for each cytokine panel (place label at top-right)  
anno_positions <- analysis_data %>%  
  group_by(cytokine) %>%  
  summarise(  
    y_max = max(log_conc, na.rm = TRUE),  
    y_range = diff(range(log_conc, na.rm = TRUE)),  
    .groups = "drop"  
  ) %>%  
  left_join(pval_anno, by = "cytokine") %>%  
  mutate(  
    x = 1.5,  
    y = y_max + y_range * 0.10  
  )  
  
# Create the boxplot  
p <- analysis_data %>%  
  ggboxplot(x = "pcos_status", y = "log_conc",
```

Figure 2. Log-transformed inflammatory-marker concentrations by PCOS status. Each panel shows one cytokine. Points represent sample-level observations, boxes summarize the group distributions, and annotations show Benjamini-Hochberg-adjusted p-values from cytokine-specific mixed-effects models that accounted for menstrual-cycle time point and repeated measures within participants.

Model used for each cytokine:

```
log1p(concentration) ~ pcos_status + time_point + (1 | pid)
```

This model tests whether cytokine concentration differs by PCOS status while adjusting for menstrual-cycle stage and accounting for repeated samples from the same participant.

Results Interpretation

Across the eight inflammatory markers measured by Luminex, no cytokine showed a statistically significant difference between PCOS and non-PCOS participants after Benjamini-Hochberg correction. The adjusted p-values shown in Figure 2 indicate that PCOS status did not strongly explain inflammatory-marker variation in this dataset after accounting for menstrual-cycle stage.

Conclusion

Conclusion: In this dataset, inflammatory marker levels did not differ significantly between participants with PCOS and participants without PCOS after adjusting for menstrual-cycle time point. The analysis correctly accounts for the repeated-measures structure by including participant ID as a random effect.